

HOME HEALTH QUALITY PERFORMANCE ANALYTICS

Medicare Home Health Patient Experience | Healthcare Operations & Executive Analytics

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Technologies	Python, Pandas, NumPy, Scikit-learn, Matplotlib, Tableau
Dataset	Medicare Home Health patient survey state data (10 states)
Report purpose	Portfolio presentation and management-level project summary

This report communicates the project's analytical workflow, findings, exploratory model results, executive dashboard, limitations, and recommended next steps.

Executive Summary

This project analyzes Medicare Home Health patient-experience measures to evaluate state-level quality performance and identify operational signals associated with overall patient satisfaction. The workflow combines data-quality review, exploratory analysis, correlation analysis, exploratory Random Forest modeling, leave-one-state-out sensitivity testing, operational scenario modeling, and Tableau business intelligence.

Across the analysis, Communication and Professional Care consistently emerged as the strongest operational signals. Professional Care showed the strongest direct correlation with overall Patient Rating ($r = 0.982$), while Communication ranked highest on average in the Random Forest sensitivity analysis (mean feature importance = 0.312).

77% Avg. Recommendation Rate	84% Avg. Patient Rating	270,466 Completed Surveys
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Management takeaway: Patient communication is the strongest data-informed improvement priority, supported by continued attention to professional care delivery. Improving Communication to the observed 75th-percentile benchmark produced a modeled +0.70 percentage-point increase in Patient Rating; improving both Communication and Professional Care produced +0.88 points.

Project Snapshot

Observations	Quality Measures	Primary Target	Analytical Scope
10 states	7 numeric measures	Overall Patient Rating	Exploratory decision support

1. Dataset & Analytical Approach

The working dataset contains 10 state-level Medicare Home Health observations and eight columns: State Abbreviation plus seven numeric patient-experience, survey-volume, and response measures. The notebook quality review identified no missing values and no duplicate rows.

Core Measures

- Professional Care - patient perceptions of professional care delivery.
- Communication - how well home health teams communicated with patients.
- Pain Medicines & Home Safety - discussions related to medications and home safety.
- Overall Patient Rating - percentage of patients giving a 9 or 10 rating; primary analytical target.
- Recommendation Rate - percentage reporting they would recommend the home health agency.
- Completed Surveys and Response Rate - survey-volume and participation measures.

Analytical Workflow

Step	Phase	What was done
1	Data review	Checked structure, data types, missing values, duplicates, and summary statistics.
2	Exploratory analysis	Compared state-level performance and relationships among quality measures.
3	Relationship analysis	Calculated Pearson correlations with overall Patient Rating.
4	Exploratory ML	Used Random Forest feature importance for operational prioritization.
5	Sensitivity testing	Repeated feature ranking while leaving out one state at a time.
6	Scenario analysis	Modeled practical 75th-percentile improvement benchmarks.
7	Executive BI	Translated results into an interactive Tableau dashboard.
Analytical scope: With only 10 aggregated state observations, this project is exploratory. The analysis supports prioritization and portfolio demonstration; it does not establish causal effects or production-ready predictive performance.		

2. Patient Satisfaction Drivers

Correlation analysis was used to identify measures most strongly associated with overall Patient Rating. Professional Care demonstrated the strongest direct operational relationship, followed closely by Communication. Recommendation Rate is interpreted separately because it is another patient-experience outcome rather than an operational care process.

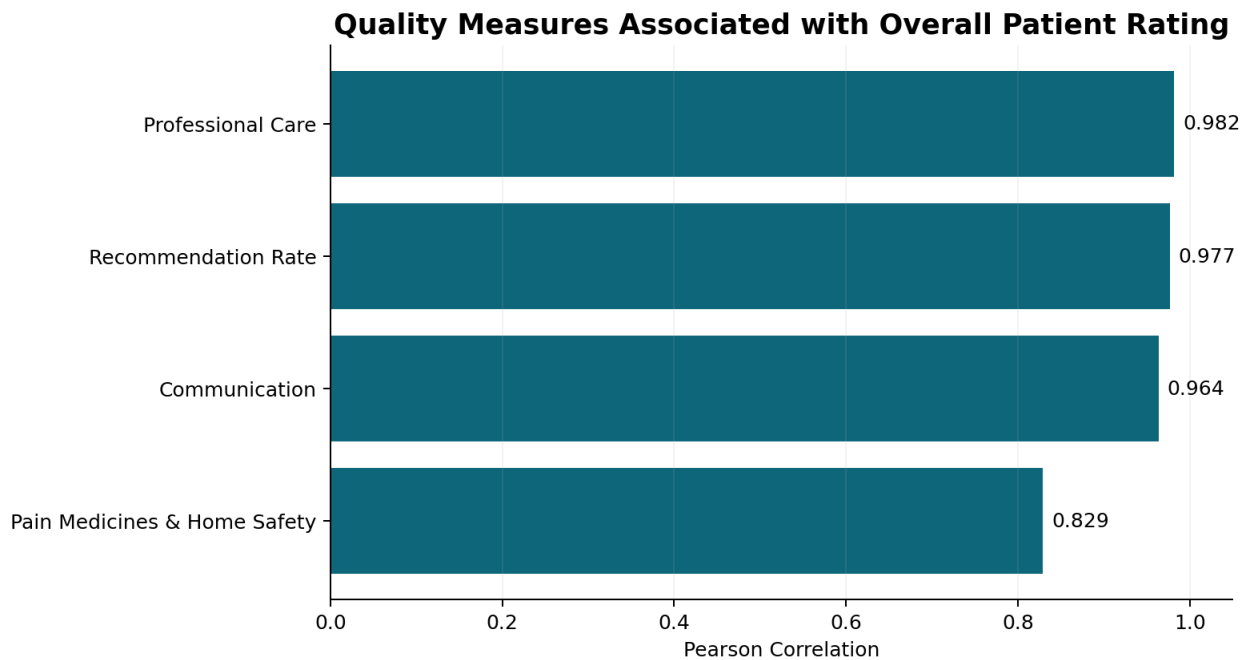


Figure 1. Correlation of selected quality measures with overall Patient Rating.

Measure	Pearson r	Interpretation
Professional Care	0.982	Strongest direct operational relationship
Recommendation Rate	0.977	Strong related patient-experience outcome
Communication	0.964	Very strong operational relationship
Pain Medicines & Home Safety	0.829	Positive but comparatively weaker relationship
Operational interpretation: Professional Care and Communication are the clearest operational areas for further quality-improvement evaluation in this dataset. Because several quality measures are highly correlated, these findings should be treated as associations rather than independent causal effects.		

3. Exploratory Machine Learning & Sensitivity Testing

A Random Forest Regressor was used as an exploratory feature-ranking tool. The predictors were Professional Care, Communication, Pain Medicines & Home Safety, Completed Surveys, and Response Rate. Recommendation Rate was intentionally excluded because it represents a related patient-experience outcome rather than an operational care process.

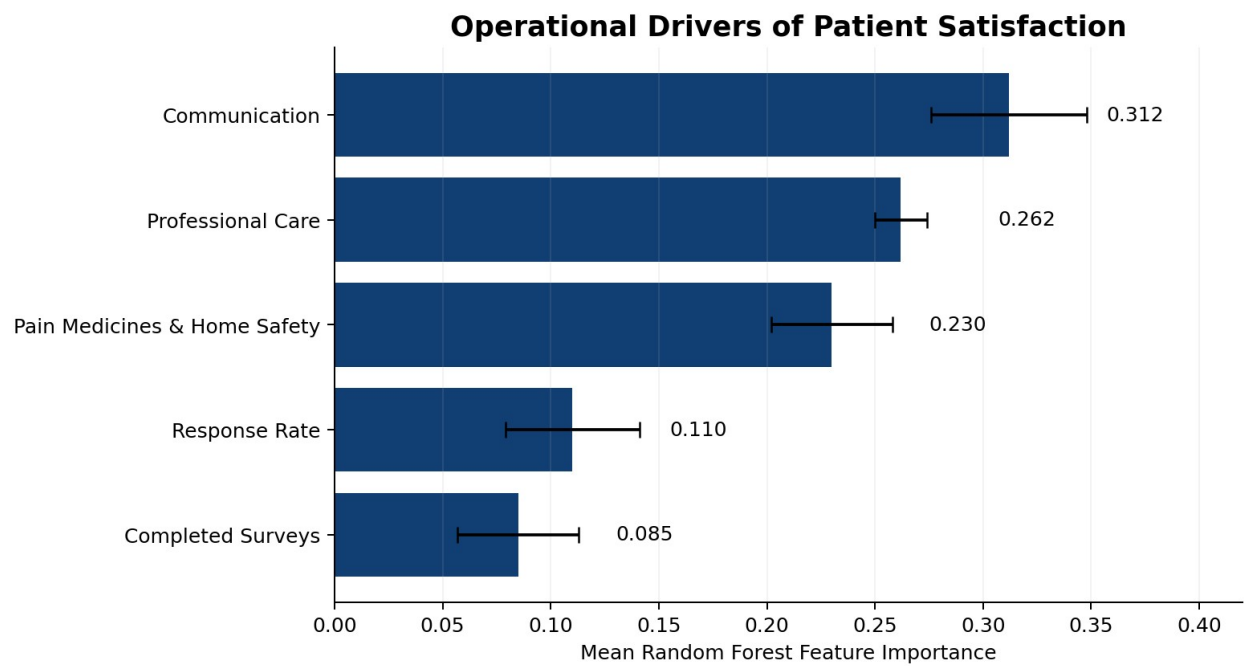


Figure 2. Mean Random Forest feature importance across leave-one-state-out iterations; error bars show standard deviation.

Feature	Mean Importance	Std. Dev.
Communication	0.312	0.036
Professional Care	0.262	0.012
Pain Medicines & Home Safety	0.230	0.028
Response Rate	0.110	0.031
Completed Surveys	0.085	0.028

Interpretation: Communication ranked highest on average, while Professional Care ranked second and showed particularly low variability across iterations (SD = 0.012). These rankings are exploratory signals, not validated predictive effects.

4. Operational Scenario Analysis

Scenario modeling estimated how Patient Rating changes when selected operational measures are moved from their average state-level values to observed 75th-percentile benchmarks. Communication increased from 84.6% to 87.5%; Professional Care increased from 87.8% to 89.0%.

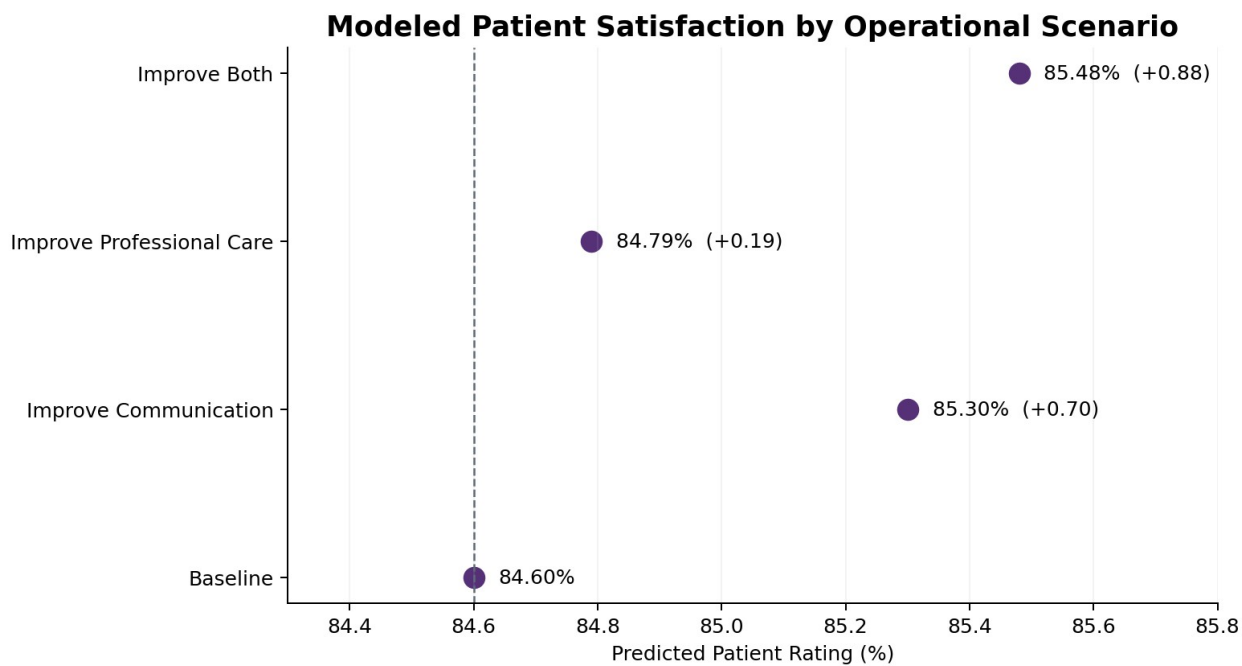


Figure 3. Modeled Patient Rating under practical operational improvement scenarios.

Scenario	Predicted Rating	Change vs. Baseline
Baseline	84.60%	-
Improve Communication	85.30%	+0.70 pp
Improve Professional Care	84.79%	+0.19 pp
Improve Both	85.48%	+0.88 pp

Executive interpretation: Communication produced the largest individual modeled improvement. Improving Communication and Professional Care together produced the strongest overall scenario. These values are planning estimates, not guaranteed intervention outcomes.

5. Executive Dashboard

The Tableau Executive Dashboard extends the analysis into a management-facing monitoring view. It combines headline KPIs with state-level comparisons for Patient Satisfaction, Recommendation Rate, and Professional Care, plus an Executive Insights panel summarizing top performers, improvement opportunities, and business impact.

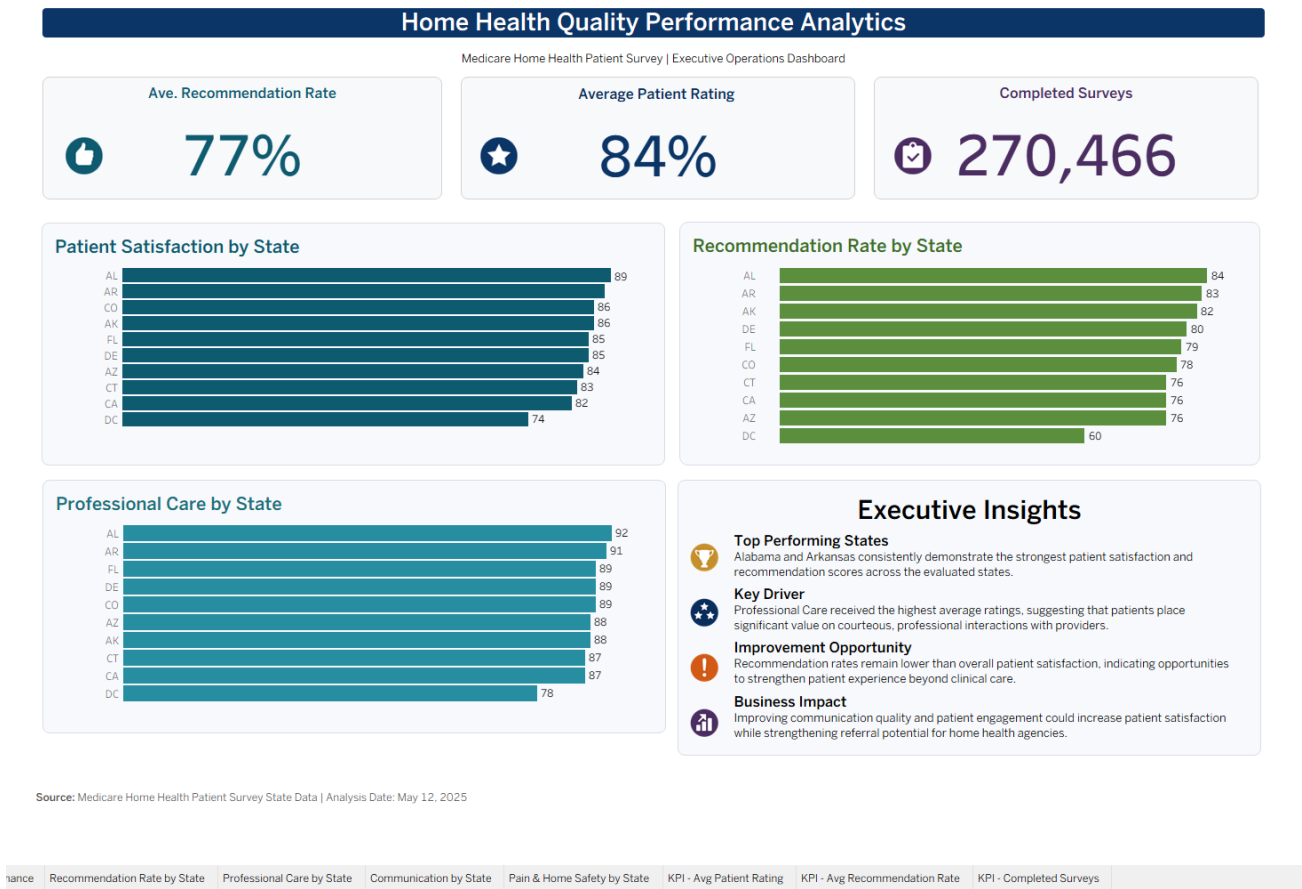


Figure 4. Home Health Quality Performance Analytics - Tableau Executive Operations Dashboard.

Dashboard Highlights

- Average Recommendation Rate: 77%.
- Average Patient Rating: 84%.
- Completed Surveys: 270,466.
- Alabama and Arkansas demonstrate the strongest satisfaction and recommendation performance among the evaluated states.
- Recommendation rates trail overall patient satisfaction, highlighting a broader patient-experience opportunity.

6. Business Recommendations

The combined evidence supports a focused quality-improvement strategy centered on Communication while maintaining strong Professional Care delivery. The recommendations below are priorities for further evaluation, not causal prescriptions.

Priority	Management Focus	Why it matters
1	Strengthen patient communication	Highest average RF importance; largest individual scenario gain (+0.70 pp).
2	Sustain professional care standards	Strongest direct correlation with Patient Rating ($r = 0.982$).
3	Coordinate both improvement areas	Strongest combined scenario: 85.48% (+0.88 pp).
4	Monitor Recommendation Rate	Related experience outcome that remains below overall Patient Rating.
5	Expand data for validation	More states, periods, and agency-level data would improve confidence and generalizability.

Limitations & Responsible Interpretation

- Small sample size: only 10 state-level observations.
- State-level aggregation should not be generalized to individual patients or agencies.
- Correlation, feature importance, and scenarios do not establish causation.
- Several quality measures are highly correlated, limiting independent interpretation.
- The Random Forest is exploratory and has not been externally validated.
- Scenario estimates use observed benchmarks and do not guarantee intervention outcomes.

Responsible analytics principle: The value of this project is not a claim of production-ready prediction. It is the disciplined combination of descriptive analysis, exploratory modeling, sensitivity testing, scenario planning, business intelligence, and transparent limitations.

7. Conclusion & Portfolio Value

This project demonstrates how healthcare quality data can be transformed into operational decision support through exploratory analysis, statistical relationships, machine learning, sensitivity testing, scenario modeling, and business intelligence.

Across these methods, Communication and Professional Care consistently emerged as the primary operational areas warranting further evaluation for patient-experience improvement. Professional Care showed the strongest direct correlation with Patient Rating, while Communication ranked highest in the sensitivity analysis and produced the largest individual modeled improvement.

The Tableau dashboard completes the workflow by translating the analysis into an executive monitoring layer. Together, the notebook, dashboard, report, and repository demonstrate the ability to connect technical analysis with healthcare operations and management communication.

Project Deliverables

Deliverable	Purpose
Jupyter Notebook	Documents data review, relationship analysis, exploratory ML, sensitivity testing, scenarios, and executive findings.
Tableau Workbook	Provides interactive state-level performance monitoring and executive visualization.
Professional Report	Communicates evidence, interpretation, recommendations, and limitations in management-ready form.
GitHub README	Provides concise recruiter-facing overview and repository navigation.
Source Dataset	Supports reproducibility of the analytical workflow.
Final takeaway: Healthcare analytics provides the greatest value when technical findings are translated into clear operational priorities. In this project, multiple analytical approaches converge on a consistent message: Communication quality and Professional Care delivery are the strongest operational signals to investigate when seeking patient-experience improvement.	

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